# Longitudinal profiles of fractional and binary variables jointly modeled with cluster formation

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# Introduction

In the banking sector, these two response variables are often modeled:

Default

- Binary variable
- Usually modeled by a logistic regression
  - Assumed independence between observations
  - Covariates measured at one point in time
  - It is used for prediction

Credit card utilization ratio

- Mixed fractional variable
  - Takes values between [0,1]
  - Has a discrete and continuous component
- It is measured longitudinally
  - The trend/slope is important
  - The shape is important
  - Behavioural information[1]
- Often aggregated or summarized[2]

The interest is in jointly modeling both variables.

[1] Baesens, B., D. Roesch, and H. Scheule (2016). *Credit risk analytics: Measurement techniques, applications, and examples in SAS*. John Wiley & Sons,

[2] Siddiqi, N. (2012). *Credit risk scorecards: developing and implementing intelligent credit scoring*. John Wiley & Sons.

# Introduction

Comparison between publications that motivated this work

| **Characteristics** | **Gaskins**[3] | **Fernández**[4] | **Proposal** |
|---|---|---|---|
| Longitudinal data | X | X | X |
| Fractional mixed variables | | X | X |
| Classification | X | | X |
| Use of covariates | | X | X |
| Cluster formation | X | | X |

[3] Gaskins, J. T., Fuentes, C. and De la Cruz, R. D. (2017). *A bayesian nonparametric model for predicting pregnancy outcomes using longitudinal profiles.*

[4] Fernandez, R., Bayes, C. L. and Valdivieso, L. (2018). *A beta-inflated mean regression model with mixed effects for fractional response variables*, Journal of Statistical Computation and Simulation **88**(10): 1936-1957.

# The beta distribution

In order to model variables restricted to the interval $(0, 1)$ a beta distribution can be used. The pdf of a random variable $Y$ that follows a beta distribution is given by

$$f_Y(y \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} y^{\alpha-1}(1-y)^{\beta-1}, \quad 0 < y < 1,$$

where $\alpha > 0$ and $\beta > 0$. The mean and variance of this distribution are given by

$$\mathrm{E}(Y) = \frac{\alpha}{\alpha + \beta} \quad \text{y} \quad \mathrm{Var}(Y) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

# The mean-precision beta distribution

In order to propose a beta regression and interpret directly the effect of covariates on the mean, Ferrari and Cribari-Neto (2004)[5]proposed the following parametrization

$$\mu = \frac{\alpha}{\alpha+\beta} \quad \text{y} \quad \phi = \alpha + \beta.$$

Then, the pdf is given by

$$b(y \mid \mu, \phi) = \frac{\Gamma(\phi)}{\Gamma(\mu\phi)\Gamma((1-\mu)\phi)} y^{\mu\phi-1}(1-y)^{(1-\mu)\phi-1}, \quad 0 < y < 1, \tag{1}$$

where $0 < \mu < 1$ and $\phi > 0$. The mean and variance are given by

$$\mathrm{E}(Y) = \mu \quad \text{y} \quad \mathrm{Var}(Y) = \frac{\mu(1-\mu)}{1+\phi}.$$

[5]Ferrari, S. and Cribari-Neto, F. (2004). *Beta regression for modelling rates and proportions*, Journal of Applied Statistics **31**: 799-815.

# The beta inflated distribution

In some cases, data contains values of zero and/or one. To solve this problem, Ospina and Ferrari (2008)[6]proposed a distribution that allows us to model this cases

$$f_Y(y \mid \delta_0, \delta_1, \mu, \phi) = \begin{cases} \delta_0, & y = 0. \\ (1 - \delta_0 - \delta_1) b(y \mid \mu, \phi), & y \in (0, 1). \\ \delta_1, & y = 1, \end{cases}$$

where $P(Y = 0) = \delta_0$, $P(Y = 1) = \delta_1$ with $\delta_0 + \delta_1 \leq 1$, $\mathrm{E}(Y \mid Y \in (0, 1)) = \mu$, $\phi > 0$ and $b(y \mid \mu, \phi)$ is the beta distribution presented in (1). The mean and variance are given by

$$\mathrm{E}(Y) = \delta_1 + (1 - \delta_0 - \delta_1)\mu$$

$$\mathrm{Var}(Y) = \delta_1(1 - \delta_1) + (1 - \delta_0 - \delta_1)\left( \frac{\mu(1 - \mu)}{1 + \phi} + (\delta_0 + \delta_1)\mu^2 - 2\mu\delta_1 \right).$$

[6]Ospina, R. and Ferrari, S. L. (2008). *Inflated beta distributions*, Statistical papers **51**, 111-126.

# The beta inflated distribution

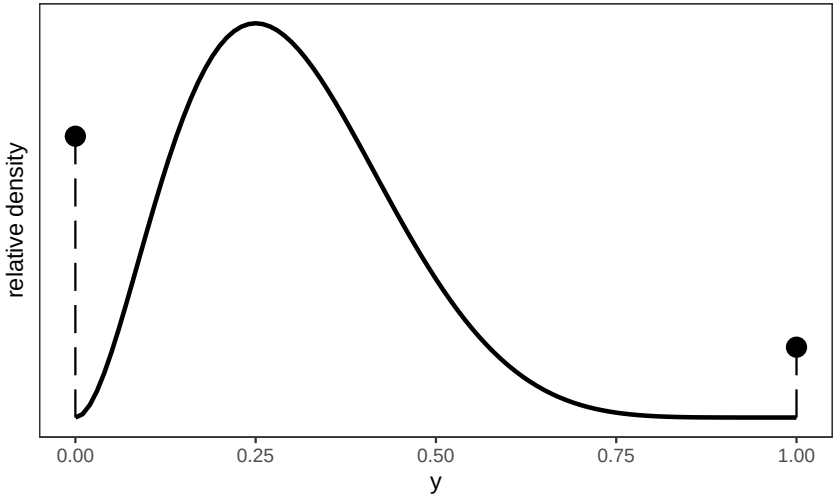



Figure: Beta inflated distribution.

# The beta inflated mean (BIm) distribution

In order to propose a beta inflated regression and interpret directly the effect of covariates on the mean, Bayes and Valdivieso (2016)[7] proposed the following parametrization

$$\gamma = \delta_1 + (1 - \delta_0 - \delta_1)\mu, \qquad \alpha_0 = \frac{\delta_0}{1-\gamma} \quad \text{and} \quad \alpha_1 = \frac{\delta_1}{\gamma}.$$

Then, the fdp is given by

$$f_Y(y \mid \alpha_0, \alpha_1, \gamma, \phi) = \begin{cases} \alpha_0(1-\gamma), & y = 0. \\ (1 - \alpha_0(1-\gamma) - \alpha_1\gamma) b(y \mid \frac{\gamma(1-\alpha_1)}{1-\alpha_0(1-\gamma)-\alpha_1\gamma}, \phi), & y \in (0,1). \\ \alpha_1\gamma, & y = 1, \end{cases} \tag{2}$$

and the mean and variance

$$\mathrm{E}(Y) = \gamma$$

$$\mathrm{Var}(Y) = \frac{(1+\alpha_1\phi)}{1+\phi}\gamma + \left( \frac{(1-\alpha_1)^2\phi}{(1-\alpha_0(1-\gamma)-\alpha_1\gamma)(1+\phi)} - 1 \right)\gamma^2.$$

[7] Bayes, C. L. and Valdivieso, L. (2016). *A beta inflated mean regression model for fractional response variables*, Journal of Applied Statistics **43**: 1814-1830.

# The model

Let $Y_i = [Y_{i1}, Y_{i2}, ..., Y_{in_i}]^\top$, $i = 1, ..., n$, be $n$ independent response mixed fractional vector variables each one observed $n_i$ times and let $D_i$, $i = 1, ..., n$ be $n$ independent binary response variables. The proposed dependent variables are modelled by:

$$\begin{aligned} D_i \mid W_i = w_i &\sim \text{Bern}(\pi_i^{w_i}) \\ Y_{ij} \mid W_i = w_i &\sim \text{BIm}(\alpha_{0ij}^{w_i}, \alpha_{1ij}^{w_i}, \gamma_{ij}^{w_i}, \phi^{w_i}) \\ W_i &\sim \text{Cat}(\boldsymbol{p})^{8}, \end{aligned} \tag{3}$$

where $W_i$ is an unobserved indicator variable of the latent class[9] $w_i$ the subject $i$ belongs to. The probability of belonging to each cluster is $\boldsymbol{p} = [p_1, p_2, ..., p_K]^\top$[10], given the total number of clusters $K$; $D_i$ is a binary variable which indicates if the subject $i$ defaults or not with probability $\pi_i^{w_i}$; $Y_{ij}$ is the fractional variable of subject $i$ at time $j$ with mean $\gamma_{ij}^{w_i}$; $\alpha_{0ij}^{w_i}$ and $\alpha_{1ij}^{w_i}$ are parameters related to $Y_{ij} = 0$ and $Y_{ij} = 1$ for subject $i$ at time $j$ and $\phi^{w_i} > 0$ is a cluster dependent precision parameter for subject $i$. Despite being unobservable, the inclusion of $W$ is important for modelling dependencies.

[8] Also known as Multinoulli

[9] Hereafter called clusters

[10] Also known as weights in the finite mixture context, where $\sum_{j=1}^{K} p_j = 1$

# The model

The proposed regression models for each parameter are:

$$\begin{aligned}
\text{logit}(\pi_i^{w_i}) &= \boldsymbol{z}_i^\top \boldsymbol{\beta}_\pi^{w_i} \\
\text{logit}(\alpha_{0ij}^{w_i}) &= \tilde{\boldsymbol{x}}_{ij}^\top \boldsymbol{\beta}_{\alpha_0}^{w_i} \\
\text{logit}(\alpha_{1ij}^{w_i}) &= \check{\boldsymbol{x}}_{ij}^\top \boldsymbol{\beta}_{\alpha_1}^{w_i} \\
\text{logit}(\gamma_{ij}^{w_i}) &= \boldsymbol{x}_{ij}^\top \boldsymbol{\beta}_\gamma^{w_i} + b_i,
\end{aligned}$$

where $\boldsymbol{\beta}_\pi^{w_i}$ is the regression parameter vector (fixed effects) for the covariates $\boldsymbol{z_i}$; $\boldsymbol{\beta}_{\alpha_0}^{w_i}$, $\boldsymbol{\beta}_{\alpha_1}^{w_i}$ and $\boldsymbol{\beta}_\gamma^{w_i}$ are the regression parameter vectors (fixed effects) for the covariate vectors $\tilde{\boldsymbol{x}}_{ij}$, $\check{\boldsymbol{x}}_{ij}$ and $\boldsymbol{x}_{ij}$, respectively, and $b_i$ is a random intercept (random effects) for the subject $i$. The random intercepts are assumed to be $b_i \sim \text{N}\left(0, \sigma_b^{2(w_i)}\right)$ i.i.d. In this proposed model, the chosen link functions are the logit function but other functions could be used in order to address for skewness or heavier tails.

# The model

The augmented likelihood function of the model is

$$L(\boldsymbol{\theta}, \boldsymbol{W}, \boldsymbol{b}) = \prod_{i=1}^{n} \prod_{k=1}^{K} \left[ p_k f_D\left(d_i \mid \boldsymbol{\theta}^w\right) \prod_{j=1}^{n_i} f_Y\left(y_{ij} \mid \boldsymbol{\theta}^w, b_i\right) \varphi\left(b_i \mid 0, \boldsymbol{\theta}^w\right) \right]^{I(w_i=k)},$$

where $Y$ and $D$, conditional on the cluster $W$, are assumed independent. Also

$$I(w_i = k) = \begin{cases} 1, & w_i = k \\ 0, & \text{otherwise,} \end{cases}$$

$p_k$ is the probability of belonging to cluster $k$, $f_D$ is the Bernoulli pmf, $f_Y$ is the BIm pdf as defined in (2) and $\varphi$ is the normal pdf.

# Bayesian inference

The augmented posterior distribution of $\boldsymbol{\theta}$, $\boldsymbol{W}$ and $\boldsymbol{b}$ can be written as

$$
\begin{aligned}
P(\boldsymbol{\theta}, \boldsymbol{W}, \boldsymbol{b}) \propto & L(\boldsymbol{\theta}, \boldsymbol{W}, \boldsymbol{b}) \times P(\boldsymbol{\theta}) \\
& \propto \prod_{i=1}^{n} \prod_{k=1}^{K} \left[ p_k f_D\left(d_i \mid \boldsymbol{\theta}^w\right) \prod_{j=1}^{n_i} f_Y\left(y_{ij} \mid \boldsymbol{\theta}^w, b_i\right) \varphi\left(b_i \mid 0, \boldsymbol{\theta}^w\right) \right]^{I(w_i=k)} \\
& \times IG(\sigma_b^2 \mid l, m) \times DIR(\boldsymbol{p} \mid \boldsymbol{u}) \\
& \times \prod_{w=1}^{K} \left[ \varphi_a(\boldsymbol{\beta}_\pi^w \mid 0, \boldsymbol{A}) \times \varphi_b(\boldsymbol{\beta}_{\alpha_0}^w \mid 0, \boldsymbol{B}) \times \varphi_c(\boldsymbol{\beta}_{\alpha_1}^w \mid 0, \boldsymbol{C}) \right. \\
& \left. \times \varphi_d(\boldsymbol{\beta}_\gamma^w \mid 0, \boldsymbol{D}) \times IG(\phi^w \mid q, r) \right],
\end{aligned}
\tag{4}
$$

where $P(\boldsymbol{\theta})$ is the prior distribution. The parameters in $\boldsymbol{\theta}$ are considered independent and the hyperparameters are chosen to give non-informative priors.

# Computational implementation

Is complex to sample from $P(\boldsymbol{\theta}, \boldsymbol{W}, \boldsymbol{b})$ with the Gibbs Algorithm because neither the posterior nor its conditional distributions are associable to any known distribution. For this reason, JAGS[11]will be used in this thesis throughout the R package *rjags*[12].

The posterior samples are invariant under permutation of indices as a property of mixtures. This leads to unidentifiability, which in the Bayesian framework is known as the label-switching problem. To overcome this problem we will use the PRA[13], which consists in estimating the MAP, which is a good approximation of the posterior mode, by

$$m^* = \arg\max_{m=1,\ldots,M} \left[ P(\boldsymbol{\theta}_m, \boldsymbol{W}_m, \boldsymbol{b}_m \mid \boldsymbol{Y}, \boldsymbol{D}) \right],$$

that will act as a pivot. This is implemented in the R package *label.switching*[14].

[11] Plummer, M. (2012). *Jags: Just another gibbs sampler*. Astrophysics Source Code Library

[12] Plummer, M., A. Stukalov, M. Denwood, and M. M. Plummer (2018). *Package 'rjags'*. update 16,1

[13] Marin, J.-M., K. Mengersen, and C. P. Robert (2005). *Bayesian modelling and inference on mixtures of distributions*. Handbook of statistics 25, 459–507.

[14] Papastamoulis, P. (2015). *label. switching: An r package for dealing with the label switching problem in mcmc outputs*. arXiv preprint arXiv:1503.02271.

# Classification of new subjects

In order to classify a new subject based on his longitudinal trajectory $y_{new}$, we can marginally calculate the new subject's PD over its random effect and cluster membership by

$$P(D_{new} = 1 \mid y_{new}, \boldsymbol{\theta}) = \sum_{k=1}^{K} P(D_{new} = 1 \mid w_{new} = k, \boldsymbol{\theta}) P(w_{new} = k \mid y_{new}, \boldsymbol{\theta}),$$

that can be expanded as

$$P(D_{new} = 1 \mid y_{new}, \boldsymbol{\theta}) = \frac{\sum_{k=1}^{K} P(D_{new} = 1 \mid w_{new} = k, \boldsymbol{\theta}) P(y_{new} \mid w_{new} = k, \boldsymbol{\theta}) P(w_{new} = k \mid \boldsymbol{\theta})}{\sum_{w=1}^{K} P(y_{new} \mid w_{new} = w, \boldsymbol{\theta}) P(w_{new} = w \mid \boldsymbol{\theta})} \quad (5)$$

where

$$P(y_{new} \mid w_{new} = k, \boldsymbol{\theta}) = \int_{-\infty}^{\infty} \prod_{j=1}^{n_{new}} f_{Y_{ij}}(y_{ij} \mid \alpha_{0ij}^{k}, \alpha_{1ij}^{k}, \gamma_{ij}^{k}(b_i), \phi^{k}) \times \varphi(b_i \mid 0, \sigma_b^2) \times db_i.$$

# Simulation study

50 datasets are generated with 400 subjects and 6 longitudinal observations.

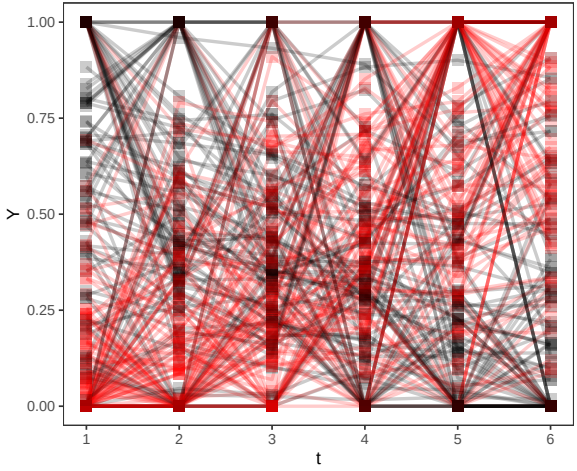



Figure: Spaghetti plot of longitudinal trajectory from a simulated dataset. The black color represents subjects that do not default, while red represents subjects that default.

# Simulation study

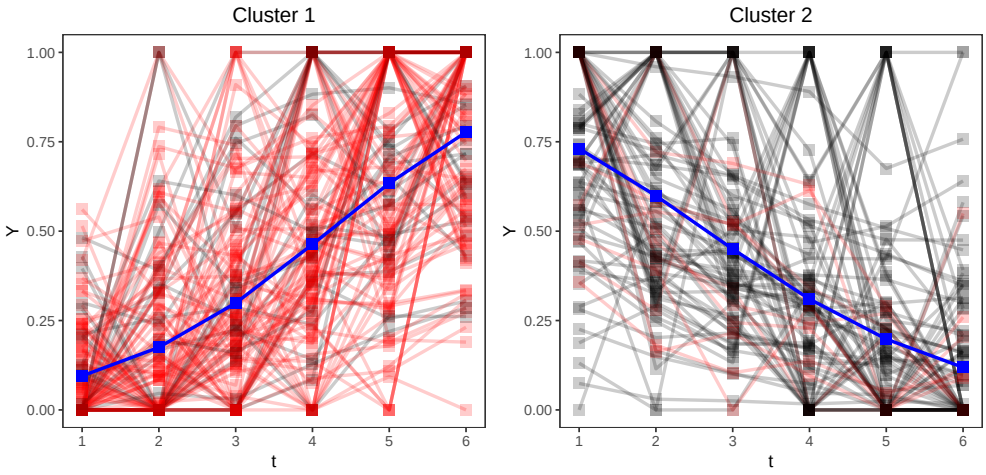



Figure: Spaghetti plots of longitudinal trajectory by cluster from a simulated dataset. The blue line is the mean $\gamma$ on each time. The black color represents subjects that do not default, while red represents subjects that default.

# Parameter recovery - Cluster 1

| Cluster | Parameter | True value | Mean | P2.5 % | P97.5 % |
|---|---|---|---|---|---|
| 1 | $\beta^1_{\alpha_0 1}$ | 0.50 | -0.51784 | -0.71421 | -0.34385 |
| | $\beta^1_{\alpha_0 2}$ | -0.80 | –0.80392 | -0.88257 | -0.73581 |
| | $\beta^1_{\alpha_1 1}$ | -1.15 | -1.14838 | -1.31197 | -0.98981 |
| | $\beta^1_{\alpha_1 3}$ | 0.60 | 0.60049 | 0.48017 | 0.74298 |
| | $\beta^1_{\gamma 1}$ | -0.80 | -0.79418 | -0.87281 | -0.71445 |
| | $\beta^1_{\gamma 2}$ | 0.90 | 0.89958 | 0.86483 | 0.93633 |
| | $\beta^1_{\gamma 3}$ | 0.80 | 0.80662 | 0.76178 | 0.86482 |
| | $\beta^1_{\pi 1}$ | -0.50 | -0.48314 | -0.90162 | -0.13216 |
| | $\beta^1_{\pi 2}$ | 2.00 | 2.05952 | 1.70876 | 2.56960 |
| | $p_1$ | 0.65 | 0.65007 | 0.60689 | 0.68786 |
| | $\phi$ | 50.00 | 49.75871 | 46.19366 | 53.16776 |

Table: Cluster 1 parameter recovery results for the proposed model (BIm).

# Parameter recovery - Cluster 2

| Cluster | Parameter | True value | Mean | P2.5 % | P97.5 % |
|---|---|---|---|---|---|
| 2 | $\beta^2_{\alpha_0 2}$ | 0.85 | 0.85772 | 0.74302 | 0.96874 |
| | $\beta^2_{\alpha_0 3}$ | -1.50 | -1.53203 | -1.68264 | -1.32170 |
| | $\beta^2_{\alpha_1 1}$ | -1.00 | -0.95881 | -1.24728 | -0.74868 |
| | $\beta^2_{\alpha_1 2}$ | -0.50 | -0.51789 | -0.61756 | -0.39027 |
| | $\beta^2_{\gamma 1}$ | -0.30 | -0.29168 | -0.40016 | -0.19491 |
| | $\beta^2_{\gamma 2}$ | -0.80 | -0.80136 | -0.85558 | -0.75881 |
| | $\beta^2_{\gamma 3}$ | 0.85 | 0.86699 | 0.74121 | 0.94757 |
| | $\beta^2_{\pi 1}$ | -0.45 | -0.44694 | -0.76779 | -0.19615 |
| | $\beta^2_{\pi 2}$ | 1.10 | 1.12601 | 0.79993 | 1.45874 |
| | $p_2$ | 0.35 | 0.34993 | 0.31214 | 0.39311 |
| | $\sigma^2_b$ | 0.50 | 0.48275 | 0.46068 | 0.50718 |

Table: Cluster 2 parameter recovery results for the proposed model (BIm).

# Parameter recovery estimation comparison

The parameter recovery estimation comparison is made against the BTran model. This model is a variation of the BIm model (3), obtained by using the transformation proposed Smithson and Verkuilen (2006)[15], which consist on transforming the fractional response variable $y' = [y(n-1) + 1/2]/n$ to constrain $y'$ to the open interval $(0, 1)$. The classification beta transformed mixed regression model with cluster formation (BTran) is given by

$$\begin{aligned} D_i \mid W_i = w_i &\sim \text{Bern}(\pi_i^{w_i}) \\ Y'_{ij} \mid W_i = w_i &\sim \text{Beta}(\mu_{ij}^{w_i}, \phi^{w_i}) \\ W_i &\sim \text{Cat}(\boldsymbol{p}). \end{aligned} \tag{6}$$

where

$$\begin{aligned} \text{logit}(\pi_i^{w_i}) &= \boldsymbol{z}_i^\top \boldsymbol{\beta}_\pi^{w_i} \\ \text{logit}(\mu_{ij}^{w_i}) &= \boldsymbol{x}_{ij}^\top \boldsymbol{\beta}_\mu^{w_i} + b_i. \end{aligned}$$

[15] Smithson, M. and Verkuilen, J. (2006). *A better lemon squeezer? maximum-likelihood regression with beta-distributed dependent variables*, Psychological methods **11**: 54.

# Parameter recovery estimation comparison - Cluster 1

| n | Cluster | Param. | True Value | BIm | BTran | P2.5 % | P97.5 % |
|---|---|---|---|---|---|---|---|
| 400 | 1 | $\beta^1_{\gamma 1}$ | -0.80 | -0.79418 | -0.39860 | -0.44074 | -0.34905 |
| | | $\beta^1_{\gamma 2}$ | 0.90 | 0.89958 | 0.56650 | 0.54105 | 0.60227 |
| | | $\beta^1_{\gamma 3}$ | 0.80 | 0.80662 | 0.25003 | 0.20158 | 0.29625 |
| | | $\beta^1_{\pi 1}$ | -0.50 | -0.48314 | -0.49085 | -0.90808 | -0.15779 |
| | | $\beta^1_{\pi 2}$ | 2.00 | 2.05952 | 2.07109 | 1.72739 | 2.54600 |
| | | $\phi$ | 50.00 | 49.75871 | 1.04928 | 1.00410 | 1.11807 |

Table: Cluster 1 parameter estimation comparison between the proposed model (BIm) and Beta Transformed (BTran) model for population size of 400.

# Parameter recovery estimation comparison - Cluster 2

| n | Cluster | Param. | True Value | BIm | BTran | P2.5 % | P97.5 % |
|---|---|---|---|---|---|---|---|
| 400 | 2 | $\beta^2_{\gamma 1}$ | -0.30 | -0.29168 | -0.13482 | -0.20560 | -0.07330 |
| | | $\beta^2_{\gamma 2}$ | -0.80 | -0.80136 | -0.51788 | -0.54762 | -0.47545 |
| | | $\beta^2_{\gamma 3}$ | 0.85 | 0.86699 | 0.66568 | 0.58172 | 0.74795 |
| | | $\beta^2_{\pi 1}$ | -0.45 | -0.44694 | -0.44103 | -0.75414 | -0.17656 |
| | | $\beta^2_{\pi 2}$ | 1.10 | 1.12601 | 1.13380 | 0.80146 | 1.50725 |
| | | $\phi$ | 50.00 | 49.75871 | 1.04928 | 1.00410 | 1.11807 |

Table: Cluster 2 parameter estimation comparison between the proposed model (BIm) and Beta Transformed (BTran) model for population size of 400.

In both clusters, none of the BTran credible intervals include the parameter's true value. The precision parameter $\phi$ is the most affected parameter by this transformation, which changes the shape of the distribution.

# Beta distribution

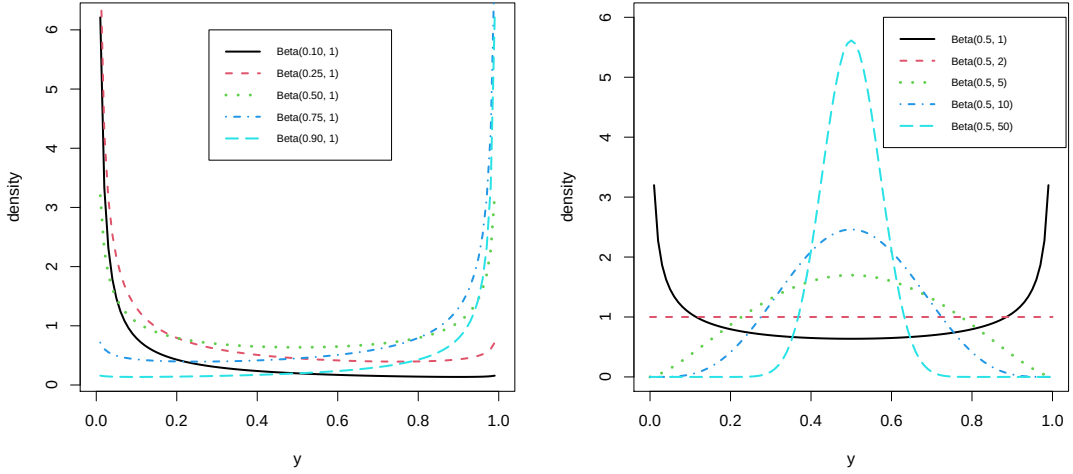



Figure: Beta distribution with the mean-variance parametrization. The notation is Beta$(\mu, \phi)$.

# Predictive power - ZOIB

For the predictive power, 4 models are compared: BIm, BTran, ZOIB, LR.

The Zero One Inflated Beta regression model (ZOIB) proposed by Liu and Kong (2015)[16]consists on the following parametrization of the beta inflated distribution:

$$f_Y(y \mid p, q, \mu, \phi) = \begin{cases} p, & y = 0. \\ (1-p)(1-q)b(y \mid \mu, \phi), & y \in (0,1). \\ (1-p)q, & y = 1, \end{cases} \tag{7}$$

where $P(Y = 0) = p$, $P(Y = 1) = (1-p)q$, $\mathrm{E}(Y \mid Y \in (0,1)) = \mu$, $\phi > 0$ and $b(y \mid \mu, \phi)$ is the beta distribution as presented in (1). This parametrization removes the constrains of the Ospina and Ferrari (2010) parametrization, which is a great advantage especially for computational simulation.

[16]Liu, F. and Kong, Y. (2015). zoib: An R Package for Bayesian Inference for Beta Regression and Zero/One Inflated Beta Regression, *The R Journal* **7**(2): 34-51.

# Predictive power - LR

The logistic regression model assumes that the observations are independent, but the longitudinal information $Y$ is not independent since it is related to the same subject.
In order to not lose longitudinal information on the application of the LR model, two extra covariates have been included. This covariates are focused on recovering the trend of $Y$ and the criteria for creating them are the same as those used by risk modeling specialists when dealing with these type of variables. The first extra covariate is

$$z_{i,3} = \begin{cases} 1, & y_{i,6} > y_{i,1}. \\ 0, & y_{i,6} = y_{i,1}. \\ -1, & y_{i,6} < y_{i,1}, \end{cases}$$

The second extra covariate is

$$z_{i,4} = \sum_{j=1}^{5} I(y_{i,j+1} > y_{i,j}) - \sum_{j=1}^{5} I(y_{i,j+1} < y_{i,j}),$$

# Predictive power

| Model | Within Sample | Out of Sample | |
|---|---|---|---|
| | AUC | Mean (AUC) | SD (AUC) |
| BIm | 0.8268 | 0.8468 | 0.0594 |
| BTran | 0.8217 | 0.8459 | 0.0595 |
| ZOIB | 0.8149 | 0.8466 | 0.0602 |
| LR | 0.8183 | 0.7325 | 0.0885 |

Table: AUC comparison between BIm, BTran, ZOIB and logistic regression models.

# Real data

| Variable | | Default | |
|---|---|---|---|
| | | 0 | 1 |
| Number of months with cash advance | Mean | 0.06 | 0.16 |
| | Median | 0.00 | 0.00 |
| Number of months with CC | Mean | 55 | 42 |
| | Median | 40 | 35 |
| Credit line | Mean | 10,373 | 7,345 |
| | Median | 5,850 | 3,300 |
| CCUR | Mean | 0.30 | 0.35 |
| | Median | 0.11 | 0.18 |

Table: Mean and median of each variable grouped by default status.

# Model structure - Linear

The following regression models the probability of default for each client ($\pi$) and the parameters associated with the CCUR ($\alpha_0, \alpha_1, \gamma$) for each client at each time.

$$
\begin{aligned}
\text{logit}(\pi_i^{w_i}) =& \beta_{\pi 0}^{w_i} + \beta_{\pi 1}^{w_i} \cdot \textit{cash_adv_total} \\
&+ \beta_{\pi 2}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\pi 3}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\alpha_{0ij}^{w_i}) =& \beta_{\alpha_0 0}^{w_i} + \beta_{\alpha_0 1}^{w_i} \cdot j + \beta_{\alpha_0 2}^{w_i} \cdot \textit{cash_adv_month} \\
&+ \beta_{\alpha_0 3}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\alpha_0 4}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\alpha_{1ij}^{w_i}) =& \beta_{\alpha_1 0}^{w_i} + \beta_{\alpha_1 1}^{w_i} \cdot j + \beta_{\alpha_1 2}^{w_i} \cdot \textit{cash_adv_month} \\
&+ \beta_{\alpha_1 3}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\alpha_1 4}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\gamma_{ij}^{w_i}) =& \beta_{\gamma 0}^{w_i} + \beta_{\gamma 1}^{w_i} \cdot j + \beta_{\gamma 2}^{w_i} \cdot \textit{cash_adv_month} \\
&+ \beta_{\gamma 3}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\gamma 4}^{w_i} \cdot \textit{credit_line} + b_i.
\end{aligned}
\tag{8}
$$

Regression structure (8) asummes that CCUR mean ($\gamma$) can only rise ($\beta_{\gamma 1} > 0$), fall ($\beta_{\gamma 1} < 0$) or stay constant ($\beta_{\gamma 1} = 0$) in time, with an extra fluctuation given by $\beta_{\gamma 2}$ (expected to be positive)

# Model structure - Dummy

$$
\begin{aligned}
\text{logit}(\pi_i^{w_i}) =& \beta_{\pi 0}^{w_i} + \beta_{\pi 1}^{w_i} \cdot \textit{cash_adv_total} + \beta_{\pi 2}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\pi 3}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\alpha_{0ij}^{w_i}) =& \beta_{\alpha_0 0}^{w_i} + \beta_{\alpha_0 1}^{w_i} \cdot I(j=1) + \beta_{\alpha_0 2}^{w_i} \cdot I(j=2) + ... \\
& + \beta_{\alpha_0 9}^{w_i} \cdot I(j=9) + \beta_{\alpha_0 10}^{w_i} \cdot I(j=10) + \beta_{\alpha_0 11}^{w_i} \cdot I(j=11) \\
& + \beta_{\alpha_0 12}^{w_i} \cdot \textit{cash_adv_month} + \beta_{\alpha_0 13}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\alpha_0 14}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\alpha_{1ij}^{w_i}) =& \beta_{\alpha_1 0}^{w_i} + \beta_{\alpha_1 1}^{w_i} \cdot I(j=1) + \beta_{\alpha_1 2}^{w_i} \cdot I(j=2) + ... \\
& + \beta_{\alpha_1 9}^{w_i} \cdot I(j=9) + \beta_{\alpha_1 10}^{w_i} \cdot I(j=10) + \beta_{\alpha_1 11}^{w_i} \cdot I(j=11) \\
& + \beta_{\alpha_1 12}^{w_i} \cdot \textit{cash_adv_month} + \beta_{\alpha_1 13}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\alpha_1 14}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\gamma_{ij}^{w_i}) =& \beta_{\gamma 0}^{w_i} + \beta_{\gamma 1}^{w_i} \cdot I(j=1) + \beta_{\gamma 2}^{w_i} \cdot I(j=2) + ... \\
& + \beta_{\gamma 9}^{w_i} \cdot I(j=9) + \beta_{\gamma 10}^{w_i} \cdot I(j=10) + \beta_{\gamma 11}^{w_i} \cdot I(j=11) \\
& + \beta_{\gamma 12}^{w_i} \cdot \textit{cash_adv_month} + \beta_{\gamma 13}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\gamma 14}^{w_i} \cdot \textit{credit_line} + b_i,
\end{aligned}
$$

# Model structure - Splines

$$
\begin{aligned}
\text{logit}(\pi_i^{w_i}) =& \beta_{\pi 0}^{w_i} + \beta_{\pi 1}^{w_i} \cdot \textit{cash_adv_total} + \beta_{\pi 2}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\pi 3}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\alpha_{0ij}^{w_i}) =& \beta_{\alpha_0 0}^{w_i} + \beta_{\alpha_0 1}^{w_i} \cdot \textit{spline_basis}_{1j} + \beta_{\alpha_0 2}^{w_i} \cdot \textit{spline_basis}_{2j} + \ldots \\
& + \beta_{\alpha_0 5}^{w_i} \cdot \textit{spline_basis}_{5j} + \beta_{\alpha_0 6}^{w_i} \cdot \textit{cash_adv_month} \\
& + \beta_{\alpha_0 7}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\alpha_0 8}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\alpha_{1ij}^{w_i}) =& \beta_{\alpha_1 0}^{w_i} + \beta_{\alpha_1 1}^{w_i} \cdot \textit{spline_basis}_{1j} + \beta_{\alpha_1 2}^{w_i} \cdot \textit{spline_basis}_{2j} + \ldots \\
& + \beta_{\alpha_1 5}^{w_i} \cdot \textit{spline_basis}_{5j} + \beta_{\alpha_1 6}^{w_i} \cdot \textit{cash_adv_month} \\
& + \beta_{\alpha_1 7}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\alpha_1 8}^{w_i} \cdot \textit{credit_line} \\
\text{logit}(\gamma_{ij}^{w_i}) =& \beta_{\gamma 0}^{w_i} + \beta_{\gamma 1}^{w_i} \cdot \textit{spline_basis}_{1j} + \beta_{\gamma 2}^{w_i} \cdot \textit{spline_basis}_{2j} + \ldots \\
& + \beta_{\gamma 5}^{w_i} \cdot \textit{spline_basis}_{5j} + \beta_{\gamma 6}^{w_i} \cdot \textit{cash_adv_month} \\
& + \beta_{\gamma 7}^{w_i} \cdot \textit{mean_age_cc} + \beta_{\gamma 8}^{w_i} \cdot \textit{credit_line} + b_i.
\end{aligned}
$$

The same regression structures are replicated for the ZOIB model.

# Goodness of fit

| K | DIC (BIm linear) | DIC (BIm dummy) | DIC (BIm spline) |
|---|---|---|---|
| 1 | 1,957 | 1,974 | 1,956 |
| 2 | **1,621** | **1,652** | **1,603** |
| 3 | 1,914 | 2,295 | 1,993 |
| 4 | 1,624 | 3,167 | 2,063 |
| 5 | 1,900 | 3,037 | 2,317 |
| 6 | 1,920 | 3,262 | 2,123 |

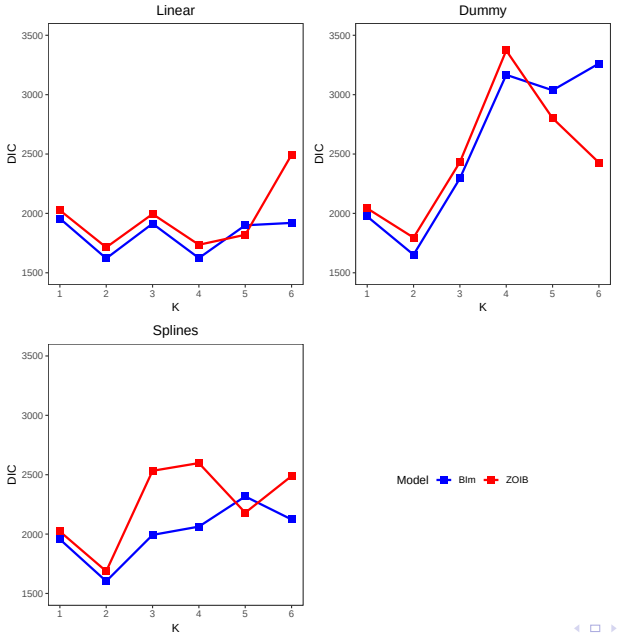
Table: DIC comparison between different number of clusters ($K$) and regression structures for the BIm model. Bold numbers indicate the best fit for each type of regression.

# Goodness of fit

| K | DIC (ZOIB linear) | DIC (ZOIB dummy) | DIC (ZOIB spline) |
|---|---|---|---|
| 1 | 2,026 | 2,042 | 2,022 |
| 2 | **1,714** | **1,796** | **1,687** |
| 3 | 1,995 | 2,432 | 2,533 |
| 4 | 1,736 | 3,372 | 2,598 |
| 5 | 1,820 | 2,802 | 2,180 |
| 6 | 2,494 | 2,428 | 2,489 |

Table: DIC comparison between different number of clusters ($K$) and regression structures for the ZOIB model. Bold numbers indicate the best fit for each type of regression.

# Goodness of fit



# Results - Linear

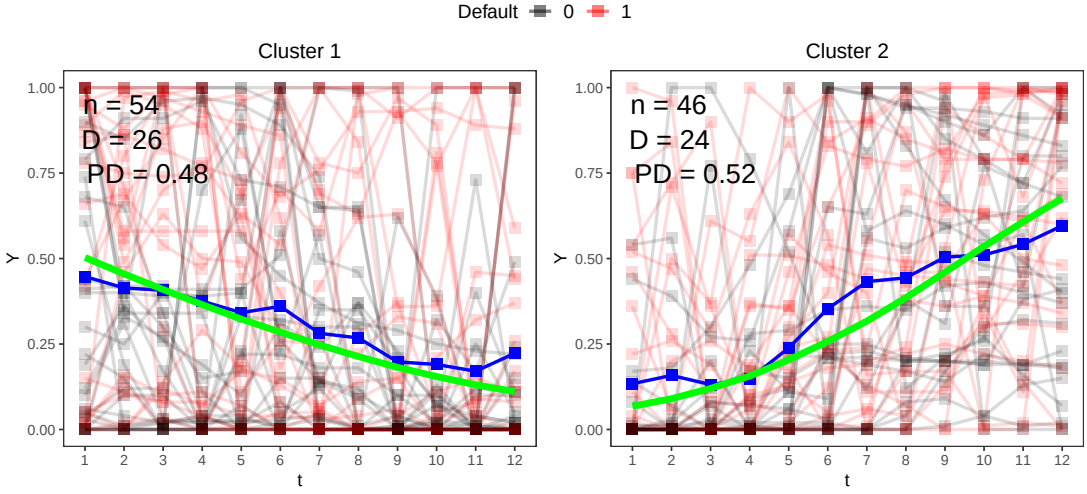



Figure: Longitudinal trajectory by cluster of real data using BIm linear regression. The blue line is the mean $\gamma$ and the green line is the estimation. $n$ is the number of clients in the cluster, $D$ is the number of default clients and $PD$ is the ratio between default and total clients.

# Results - Dummy

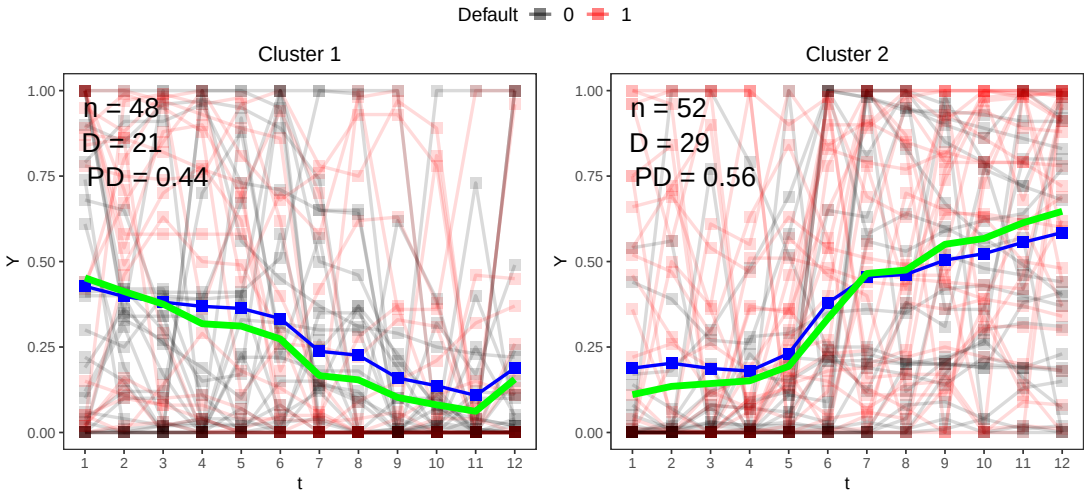



Figure: Longitudinal trajectory by cluster of real data using BIm dummy regression. The blue line is the mean $\gamma$ and the green line is the estimation. $n$ is the number of clients in the cluster, $D$ is the number of default clients and $PD$ is the ratio between default and total clients.

# Results - Splines

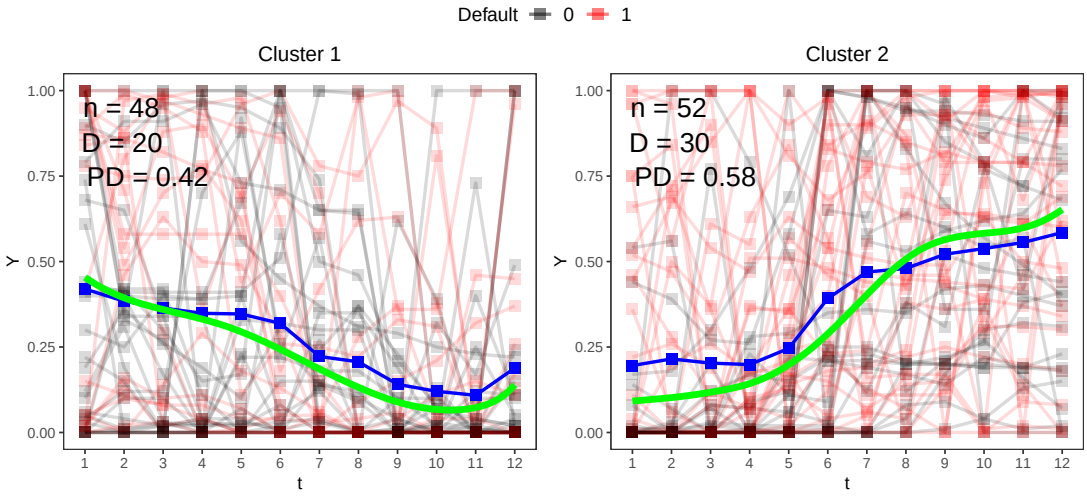



Figure: Longitudinal trajectory by cluster of real data using BIm spline regression. The blue line is the mean $\gamma$ and the green line is the estimation. $n$ is the number of clients in the cluster, $D$ is the number of default clients and $PD$ is the ratio between default and total clients.

# Fixed effects estimation

| Cluster | Parameter | Covariate | Mean | P 2.5 % | P 97.5 % |
|---|---|---|---|---|---|
| 1 | $\beta_{\gamma 7}$ | *mean_age_cc* | 0.20794 | 0.03277 | 0.39066 |
| | $\beta_{\gamma 8}$ | *credit_line* | -0.07006 | -0.11132 | -0.02140 |
| | $\beta_{\pi 1}$ | *cash_adv_total* | 0.32887 | 0.05010 | 0.71331 |
| 2 | $\beta_{\gamma 6}$ | *cash_adv_month* | 0.35003 | 0.01508 | 0.64164 |
| | $\beta_{\gamma 7}$ | *mean_age_cc* | -0.16484 | -0.27058 | -0.07967 |
| | $\beta_{\pi 2}$ | *mean_age_cc* | -0.33886 | -0.64876 | -0.08421 |

Table: Estimated posterior distribution of parameters from BIm spline regression with 2 clusters applied to real data.

# Prediction power

| Model | Regression | K | AUC Out of sample |
|---|---|---|---|
| BIm | Linear | 2 | 0.6128 |
| | Dummy | 2 | 0.6036 |
| | Splines | 2 | 0.6272 |
| ZOIB | Linear | 2 | 0.6224 |
| | Dummy | 2 | 0.6388 |
| | Splines | 2 | 0.6308 |
| LR | | 1 | 0.6076 |

Table: AUC comparison between BIm, ZOIB and LR models.

# Conclusions

- Simulation study
  - BTran model induces bias in the parameters related to longitudinal information
  - BIm, BTran and ZOIB models outperformed LR model in out-of-sample prediction power
- Application
  - ZOIB and BIm regressions outperformed the LR model in out-of-sample prediction
  - Cluster formation facilitates interpretation of client behavior profiles and favours the inference of a default status
- Suggestions for future studies
  - Using the Flexible Beta[17](FB) regression which is robust for bi-modal fractional distributions
  - Reduce the number of dummy variables in the dummy regression using the effect fusion model-based clustering[18]
  - Using the Tobit-like[19]models restricted to [0, 1] that support any distribution in the interval

[17] Migliorati, S., Di Brisco, A. M. and Ongaro, A. (2018). *A new regression model for bounded responses*, Bayesian Anal. **13**(3): 845–872.

[18] Malsiner-Walli, G., Pauger, D. and Wagner, H. (2018). *Effect fusion using model-based clustering*, Statistical Modelling **18**(2): 175–196.

[19] Hossain, A., Rigby, R., Stasinopoulos, M., and Enea, M. (2016). *Centile estimation for a proportion response variable*. Statistics in medicine, **35**(6), 895-904.